

Lab Experiment: 4

Title: Application of Classification Algorithm (J48) for Decision Tree Construction

Course title: Data Mining Problem Description

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1. Objective

To apply the J48 decision tree classification algorithm on a student performance dataset, construct a pruned decision tree, and evaluate the model using 10-fold cross-validation.

2. Tools & Software

- WEKA (Waikato Environment for Knowledge Analysis)
 - Microsoft Excel / Google Sheets
 - Dataset in CSV and ARFF format
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3. Dataset Description

The dataset contains academic results of students across three academic years and their overall performance class.

Relation Name: Student_Performance

Number of Instances: 52

Number of Attributes: 5

Attribute	Description
FirstYear	GPA of first year
SecondYear	GPA of second year
ThirdYear	GPA of third year
AVG Till Now	Average GPA till current year
Performance_Class	Final performance category (target)

Target Classes: Excellent, Good, Average, Poor

4. Methodology / Procedure

1. Prepared dataset in Excel sheet.
2. Saved the dataset in CSV format.
3. Converted CSV file into ARFF format using WEKA.
4. Loaded the ARFF file into WEKA Explorer.
5. Selected **Classify** → **J48 algorithm**.
6. Used default parameters (Confidence Factor = 0.25, MinNumObj = 2).
7. Applied **10-fold cross-validation**.
8. Generated and visualized the decision tree.

9. Evaluated model performance using accuracy metrics and confusion matrix.

5. J48 Model Information

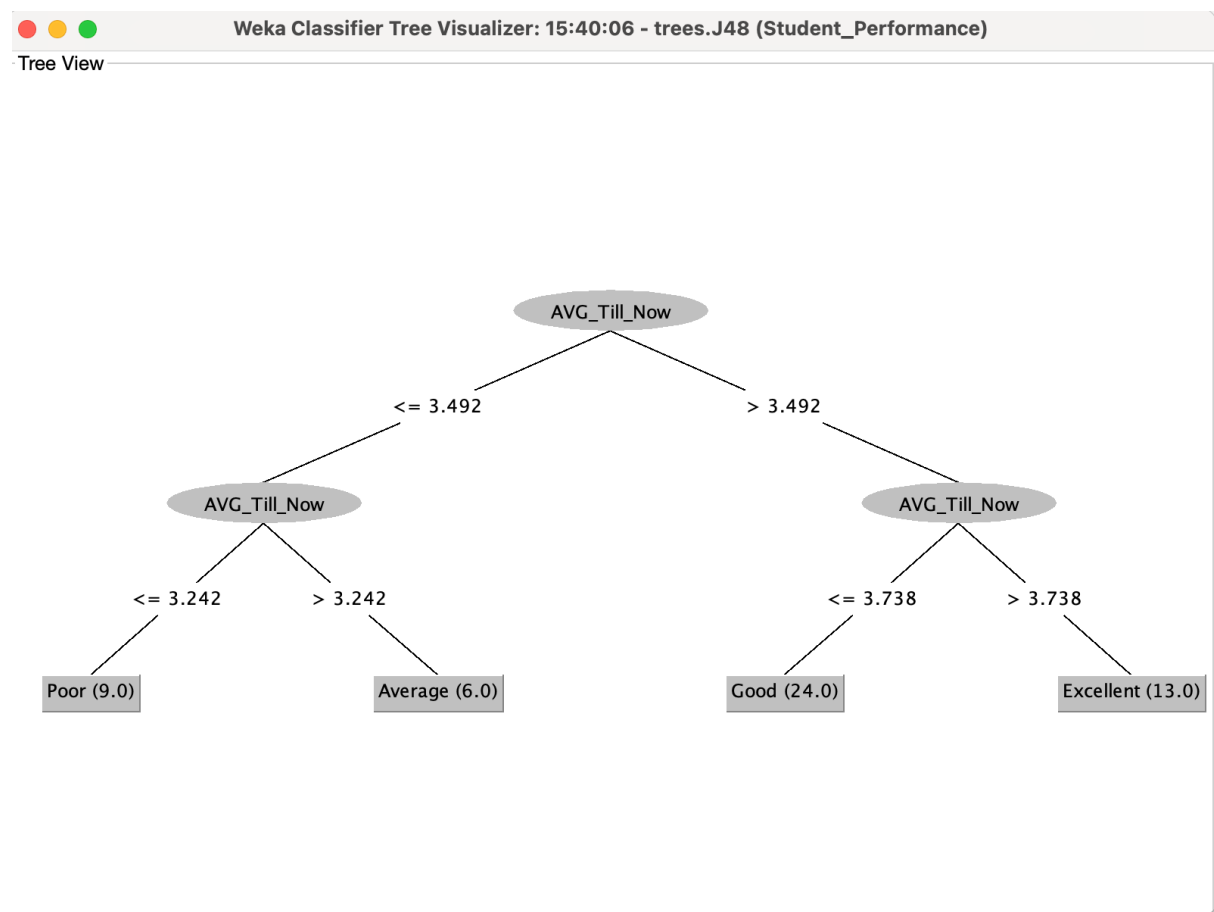
- Scheme: weka.classifiers.trees.J48
- Confidence Factor (C): 0.25
- Minimum Instances per Leaf (M): 2
- Tree Type: Pruned Tree

Tree Size: 7 nodes

Number of Leaves: 4

Time to Build Model: 0 seconds

6. Generated Decision Tree (Rules)



The tree uses only the attribute AVG_Till_Now to classify student performance.

IF $\text{AVG_Till_Now} \leq 3.242 \rightarrow \text{Performance} = \text{Poor}$

IF $3.242 < \text{AVG_Till_Now} \leq 3.492 \rightarrow \text{Performance} = \text{Average}$

IF $3.492 < \text{AVG_Till_Now} \leq 3.738 \rightarrow \text{Performance} = \text{Good}$

IF $\text{AVG_Till_Now} > 3.738 \rightarrow \text{Performance} = \text{Excellent}$

This shows that overall GPA is the most significant factor in determining performance class.

7. Evaluation Results (10-Fold Cross-Validation)

7.1 Summary Statistics

- Correctly Classified Instances: 49 (94.23%)
- Incorrectly Classified Instances: 3 (5.77%)
- Kappa Statistic: 0.9151 (Very high agreement)
- Mean Absolute Error: 0.0288
- Root Mean Squared Error: 0.1698

Interpretation: The model demonstrates excellent predictive performance.

7.2 Accuracy by Class

Class	Precision	Recall	F-Measure
Excellent	0.929	1.000	0.963
Good	0.958	0.958	0.958
Average	0.833	0.833	0.833
Poor	1.000	0.889	0.941

This table shows **how well the model predicts each performance category individually.**

- **Precision** → How many predicted cases of a class were actually correct
- **Recall** → How many actual cases of a class were correctly identified
- **F-Measure** → Overall balance between Precision and Recall

Interpretation:

- **Excellent:**

Model identified all Excellent students correctly (Recall = 1.000). Very reliable predictions.

- **Good:**

High accuracy (0.958). Only very few mistakes.

- **Average:**

Lowest performance (0.833). Some Average students were misclassified.

- **Poor:**

Perfect Precision (1.000) → Whenever the model predicts Poor, it is always correct.

Recall is slightly lower (0.889) → One Poor student was predicted as another class.

Overall: The classifier performs very well for all classes, especially Excellent and Good.

7.3 Confusion Matrix

Actual \ Predicted	Excellent	Good	Average	Poor
Excellent	13	0	0	0
Good	1	23	0	0
Average	0	1	5	0
Poor	0	0	1	8

This table shows the **exact number of correct and incorrect predictions** made by the model.

- Rows = Actual class
- Columns = Predicted class
- Diagonal values = Correct predictions
- Off-diagonal values = Errors

Interpretation:

- **Excellent:**

All 13 students correctly classified → Perfect performance.

- **Good:**

23 correct, 1 misclassified as Excellent.

- **Average:**

5 correct, 1 misclassified as Good.

- **Poor:**

8 correct, 1 misclassified as Average.

Most values are on the diagonal → The model is highly accurate (94.23%).

8. Discussion

- The J48 algorithm selected only one attribute (AVG_Till_Now) for classification.
 - This indicates that cumulative GPA strongly determines final performance.
 - The model achieved high accuracy (94.23%), showing that the dataset is well structured and separable.
 - Minor misclassifications occurred between adjacent performance categories (e.g., Good vs Excellent).
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9. Conclusion

The experiment successfully applied the J48 decision tree algorithm to classify student performance. The resulting model is simple, interpretable, and highly accurate. The study demonstrates that overall GPA is the dominant factor in predicting academic performance categories.

10. Precautions

- Ensure dataset has no missing or inconsistent values.
 - Normalize or verify GPA scale consistency.
 - Choose appropriate class labels.
 - Use cross-validation to avoid overfitting.
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11. Applications

- Academic performance prediction systems
- Student monitoring and advising tools
- Early identification of at-risk students
- Educational data mining research